FUAD AMIRGULIYEV, FRM — AI/ML SENIOR FINANCIAL RISK ANALYST

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Senior Financial Risk Analyst with over **10 years** building credit risk models and regulatory frameworks for regional banks managing $2B+ portfolios across retail, SME, corporate, and mortgage lending. Specialized in deploying **machine learning models** (Random Forest, XGBoost, Cox Proportional Hazards) using **Python, R, and SQL** for **IFRS 9, CECL compliance, and fraud detection** across 180k+ annual loan applications.

- **Credit Risk Modeling**: Built **PD/LGD/EAD models** covering 180k+ accounts with $2.1B exposure achieving **external auditor approval** on first submission using **Python, R, SQL**
- **Fraud Detection**: Deployed **network analysis systems** using **Python/NetworkX** on 1M+ monthly transactions identifying $15M+ in organized fraud rings working with fraud operations teams
- **Process Automation**: Created **Python workflows (Pandas, NumPy, Plotly)** replacing 40-hour manual provisioning with automated systems, built **Power BI dashboards** for **Risk Committee presentations** tracking 35+ KRIs
- **Model Validation**: Validated 5+ production credit scorecards using champion-challenger testing, calculated **KS statistics, Gini coefficients, PR AUC, AUC metrics** meeting **SR 11-7** regulatory standards

KEY ACHIEVEMENTS

- Built **ML credit scoring platform** for **personal loans $1k-$50k** and **SME business loans up to $500k** using Random Forest, XGBoost, LightGBM, logistic regression, and selected logistic regression as the final model on 150+ features from **Experian/Equifax bureaus**, deployed via **REST API to Finastra** working with IT team, automated 6,200 monthly approvals preventing $12M+ in defaults
- Developed **CECL lifetime loss engine** using **Cox Proportional Hazards** in **Python and R** for $1.8B portfolio, automated monthly provisioning with **Python (Pandas)** across 12 stress scenarios, presented results to **Risk Committee and Board** informing $45M capital buffer decision
- Engineered **CECL framework** for 6 product lines with 100k+ accounts and $2.1B exposure, built **PD/LGD/EAD models**, implemented **3-stage classification** monitoring delinquency triggers, partnered with CFO's finance team on monthly $25M provisions, delivered documentation achieving **external auditor approval** 8 weeks ahead of deadline
- Deployed **fraud detection system** using **network clustering (Python/NetworkX)** on 1M+ monthly transactions, identified connected accounts through shared devices and beneficiary patterns, worked with fraud operations deploying 25 triggers in **Fiserv platform**, caught **$15M in fraud schemes** with precision
- Automated **loan loss allowance reporting** using **Python** extracting data from **Temenos T24** via SQL, generated executive dashboards for Chief Risk Officer tracking NPL ratios and transition matrices with drill-down by product and geography

Expertise in IFRS 9, CECL, credit scoring, PD/LGD/EAD modeling, fraud analytics, stress testing, model validation, machine learning, Python, R, SQL, working with Risk Committees and regulatory auditors.
Authorized to work for any U.S. employer without visa sponsorship.

PROFESSIONAL EXPERIENCE

Provident Bank — **September 2022 – October 2025**
Regional bank with multi-product portfolio across retail, SME, and corporate lending, 3rd largest bank in market by assets.

SENIOR CREDIT RISK ANALYST
Worked on CECL implementation and model validation reporting to VP of Credit Risk. Supported 2 senior analysts partnering with IT and finance teams on regulatory projects.

- Managed **CECL transition project** analyzing 80k+ accounts across retail, SME, and corporate portfolios, built **forward-looking ECL models** with lifetime PD using Cox PH model, integrated GDP, unemployment, and interest rate forecasts, developed **stage migration matrices** tracking transitions, worked with IT extracting historical data for model development
- Expanded **stress testing framework** building 12 scenarios covering credit, market, and liquidity risk, partnered with IT automating execution through **Python scripts**, integrated outputs into Moody's Analytics platform for CFO and Treasurer ICAAP planning, stress results showed losses ranging $8M to $75M informing capital decisions
- Validated **15+ credit scorecards** across retail, cards, SME, and mortgage using **backtesting on out-of-time samples**, calculated KS, Gini, AUC, PSI, and calibration metrics, identified 4 scorecards with drift, recommended recalibration achieving 0.71 Gini meeting regulatory standards
- Built **portfolio segmentation framework** using **k-means clustering in Python** on borrower characteristics, created 8 risk segments showing concentration risks, developed dashboards tracking segment performance, presented to Credit Committee informing limit adjustments
- Built **credit policy simulation tool in Excel with VBA** modeling policy changes (DTI limits, LTV caps, minimum scores), analyzed 50k+ applications, estimated impact on approvals, losses, and revenue, presented scenarios to Credit Committee balancing growth vs risk

Bank Respublika — **March 2017 – September 2022**
Diversified financial institution with $2B+ in assets serving 200k+ customers across consumer, SME, and corporate banking, top-3 market position.

SENIOR FINANCIAL RISK ANALYST
Built credit risk models and fraud detection systems using **Python, R, and SQL** reporting to the Chief Risk Officer. Partnered with IT, finance, and fraud operations teams on model development and deployment.

- Developed **IFRS 9 ECL models** for **6 product lines** covering **100k+ accounts** with **$2.1B exposure**, built **PD/LGD/EAD models** using **logistic regression** and **vintage analysis**, implemented **3-stage classification** monitoring **30-day delinquency triggers**, worked with **finance team** on monthly **$85M provision calculations**, delivered regulatory documentation achieving **Central Bank approval** on first submission
- Built **ECL lifetime loss models** using **R** for **retail, SME, and corporate portfolios**, incorporated **macroeconomic forecasts** (GDP, unemployment, interest rates) from **economics team**, automated quarterly provisioning pipeline using **Python (Pandas, NumPy)**, documented methodology for **external auditors**
- Expanded **stress testing framework** from 3 to 12 scenarios using **Monte Carlo simulation in R** with **10,000 iterations** covering **credit, market, and liquidity risk**, quantified losses ranging **$25M to $180M** under adverse scenarios, partnered with **IT team** automating execution through **Python scripts**, presented results to **Risk Committee and Board** informing **capital buffer** and **dividend policy decisions**
- Validated **15+ credit scorecards** for **retail, SME, mortgage, and card products** using **champion-challenger testing**, calculated **KS statistics, Gini coefficients, AUC metrics** on **out-of-time validation samples**, measured stability using **PSI**, identified 4 scorecards requiring recalibration, worked with **modeling team** achieving **0.71 Gini** meeting **SR 11-7 regulatory standards**
- Developed **fraud detection system** using **machine learning** and **network analysis (Python/NetworkX)** on **500k+ monthly transactions**, built **behavioral features** analyzing payment patterns for "on-time payers" executing **bust-out schemes**, created **spatial features** detecting geographic anomalies, deployed **real-time rule-based triggers** working with **fraud operations team**, identified **$15M in organized fraud rings** with **91% precision rate**
- Built **customer segmentation models** using **k-means clustering in Python (scikit-learn)** analyzing **behavioral data** (payment patterns, channel usage, product holdings), created **8 risk-based segments** for targeted **underwriting policies**, developed **monitoring dashboards** tracking segment migrations and performance metrics, presented findings to **Credit Committee** informing **DTI limits, LTV caps, and minimum credit score policies**
- Automated **risk reporting suite** using **Python** extracting data from **core banking system** via **SQL**, generated **executive dashboards** for **Chief Risk Officer** tracking **NPL ratios, coverage ratios, roll rates**, and **concentration limits** with drill-down by **product type, geography, and vintage cohort** for monthly **Risk Committee meetings**
- Engineered **behavioral and spatial features** for **fraud analytics** identifying high-risk patterns (rapid **credit line increases**, **velocity spikes**, new **beneficiary concentrations**), deployed triggers in **transaction monitoring system** catching fraud earlier in customer lifecycle, worked with **IT and fraud operations teams** on production integration

AccessBank **January 2015 – February 2017**

Regional lender with $300M+ loan portfolio focused on consumer and auto lending.

RISK ANALYST

Built forecasting models and portfolio monitoring systems reporting to Senior Risk Manager. Supported risk team with quantitative analysis for loss forecasting and capital planning.

- Developed **transition matrix forecasting models** using **Markov chain analysis in Python** estimating migrations across delinquency buckets for $300M portfolio covering **personal loans $2k-$35k and auto loans $5k-$40k**, incorporated seasonal patterns and macroeconomic indicators, achieved 89% forecast accuracy enabling stable provisioning
- Built **Value-at-Risk models** using historical simulation, variance-covariance, and **Monte Carlo in R** with 10,000 iterations, calculated 99% VaR reaching $8M in severe scenarios, implemented two-stage backtesting, quantified tail risk using Expected Shortfall, informed $12M economic capital allocation and RAROC hurdle rates
- Automated **daily portfolio monitoring** using **Python and SQL** covering 50k+ accounts, created dashboards monitoring concentration limits (industry, geography, large exposures), flagged breaches with email alerts for escalation, eliminated 14 hours weekly manual work
- Analyzed **early warning indicators** tracking roll rates, cure rates, payment-to-income ratios, built **logistic regression models (Python/scikit-learn)** predicting 90+ day delinquency within 6 months with 0.76 AUC, identified $45M in high-risk exposure, worked with Collections Manager implementing early intervention programs

EDUCATION

Master of Science in Data Science, Pace University, Seidenberg School of Computer Science and Information Systems

Master of Science in Accounting and Finance, University of Essex

Bachelor of Science in Finance, Qafqaz University

PROFESSIONAL DEVELOPMENT

Financial Risk Manager (FRM), Global Association of Risk Professionals (GARP)

COMMUNITY LEADERSHIP

Risk Management Mentor helping banking professionals transition to quantitative risk roles through one-on-one sessions covering credit modeling techniques, regulatory frameworks (IFRS 9, CECL), and machine learning applications.

Guest Speaker at banking conferences on practical applications of machine learning in credit risk including model development, validation, and regulatory compliance.

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EXECUTIVE STRENGTHS

Credit Risk Modeling, Financial Risk Management, IFRS 9, Calculated Expected Credit Loss (CECL), Basel III, PD/LGD/EAD Estimation, Credit Scoring, Scorecards Development, Loss Forecasting, Allowance for Credit Losses (ACL), Model Development, Model Validation, Model Governance, Stress Testing, Scenario Analysis, Sensitivity Analysis, Fraud Analytics, Portfolio Analytics, Underwriting Strategy, Risk Reporting, Regulatory Compliance, Model Risk Management (SR 11-7), CCAR, DFAST, Capital Planning, Statistical Modeling, Machine Learning, Quantitative Analysis, Process Automation, Dashboard Development, Stakeholder Management, Risk Committee Presentations

TECHNICAL SKILLS

Python (Pandas, NumPy, scikit-learn, NetworkX, Matplotlib, Plotly), R (survival, caret), SQL, Logistic Regression, Random Forest, XGBoost, K-Means Clustering, Cox Proportional Hazards, PD/LGD/EAD Estimation, Credit Scoring, WOE/IV Binning, KS Statistics, Gini Coefficient, AUC Analysis, IFRS 9, CECL, Basel III, SR 11-7, Stress Testing, Monte Carlo Simulation, Model Validation, Power BI, Tableau, Excel (VBA), Temenos T24